

PUNEETH MUTHINENI

Software Engineer

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SUMMARY

Software engineer with close to two years of experience shipping production features end to end, plus an MS in Computer Science. Works across the stack, from React interfaces and Node.js gateways to Python/FastAPI services, REST APIs, and PostgreSQL and MongoDB persistence. Treats testing as engineering rather than cleanup: builds pytest suites, evaluation harnesses, and CI gates that catch regressions before release, and has reproduced, root-caused, and validated defects in live production systems. Builds with LLM agents in the product path using LangGraph and Claude/OpenAI APIs, wrapping them in deterministic guardrails and fallback behavior so agent outputs stay predictable and verifiable. Ships in small fast increments, documents work clearly, and learns unfamiliar systems quickly through code review and direct stakeholder feedback.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (Node.js / React)

Testing & Quality Automation: pytest, unit/integration/regression testing, CI gates and quality gates, evaluation harnesses, defect reproduction and root-cause analysis, edge-case testing, input/output validation, code review

AI Agents & LLM Tooling: LangGraph, LangChain, Anthropic Claude and OpenAI APIs, agentic workflows, prompt design, tool/function calling, structured outputs, deterministic guardrails and fallback behavior, RAG, FAISS vector search

Frontend: React, component-based UI, front-end to back-end integration

Backend & APIs: FastAPI, Flask, Node.js, REST/JSON APIs, WebSockets, microservices, Pydantic request validation, service boundaries, API contracts

Databases: PostgreSQL, MongoDB, SQL, data modeling, JSON processing

DevOps & Developer Tooling: Docker, Linux, Git/GitHub, GitHub Actions, Jenkins, CI/CD pipelines, build and lint automation, monitoring, logging, alerting

Cloud: AWS (S3, EC2), Azure, GCP

Data & ML: pandas, NumPy, data cleaning/transformation/validation, scikit-learn, PyTorch, Hugging Face Transformers, XGBoost, BERT, MLflow experiment tracking, Optuna

Practices: Agile, pull requests and code review, technical documentation, stakeholder communication, incident troubleshooting

WORK EXPERIENCE

Software Developer (Contract) - Acentle

Jan 2025 - Present

- Built, shipped, and supported backend and AI-enabled product features across 5+ production releases, owning design, implementation, testing, deployment, and post-release support with minimal supervision.
- Automated build, lint, and pytest workflows through CI/CD pipelines, improving release consistency and cutting deployment-validation time by approximately 30%.
- Diagnosed production incidents by tracing logs, alerts, API behavior, and system outputs to root cause, then validated fixes and hardened reliability across backend services.
- Applied quality and security review practices by validating inputs and outputs, testing edge cases, and reviewing generated or modified code before every release.
- Developed SyncDev Spatial, a Linux-based Python platform exposing 10+ REST/JSON API endpoints with WebSocket-based communication, Docker deployment, and monitoring/logging, reducing manual operational overhead by approximately 40%.
- Designed Python and SQL data flows to clean, transform, validate, and serve operational data for downstream application workflows and automation.
- Integrated backend services with REST APIs, JSON payloads, and SQL-backed data logic, adding monitoring workflows that improved data availability and application reliability.
- Authored documentation covering API behavior, deployment steps, troubleshooting guidance, and support processes for engineering and stakeholder teams.

Software Developer Intern - Nice Interactive

Jan 2024 - Jun 2024

- Added unit, integration, and regression tests across backend modules to improve reliability and reduce repeat defects.
- Reproduced, debugged, and resolved defects by reviewing application behavior, testing backend changes, and validating fixes before release.
- Built Python backend features and REST API integrations supporting data exchange, service functionality, and internal application workflows.

- Wrote SQL across 3+ application modules for data retrieval, reporting outputs, backend workflows, and application-level data validation.
- Shipped through Git-based workflows including feature branches, pull requests, code review, and CI/CD runs, incorporating senior-engineer feedback on readability and maintainability.
- Collaborated with developers and stakeholders in an agile environment to clarify requirements and deliver backend updates on time, documenting API behavior, SQL logic, and troubleshooting steps.

PROJECTS

CareCompass - Full-Stack AI Product with Agent Workflow

React | Node.js | Python/FastAPI | LangGraph | MongoDB | Docker | pytest

- Shipped a production-style full-stack application end to end, spanning a React UI, a Node.js service gateway, and a Python/FastAPI backend with REST APIs and request validation, owning complete features from interface through persistence.
- Integrated a LangGraph agent into the live product path, delivering AI-assisted recommendations while keeping the backend service layer responsible for validation, persistence, and application logic.
- Engineered deterministic safety guardrails and fallback paths around agent outputs so AI-assisted responses stayed predictable, testable, and monitorable in real user workflows.
- Containerized the stack with Docker and enforced pytest checks and CI gates on every change to catch regressions and sustain release confidence as features evolved.
- Documented API behavior, data flow, agent workflow logic, deployment steps, and troubleshooting guidance to support handoff across front-end, backend, and AI components.

MLOps Prediction Service - Evaluation Harness & Model Lifecycle

Python | pytest | MLflow | Optuna | scikit-learn | FastAPI | PostgreSQL

- Designed a 14-test pytest CI evaluation harness to validate model behavior, catch regressions, and enforce quality gates before deployment.
- Ran hyperparameter optimization with Optuna across 96 tracked MLflow experiments, comparing performance and evaluation metrics to improve F1 score by 3.3% and AUC by 2.9%.
- Deployed the trained model as a FastAPI and PostgreSQL inference service with Pydantic validation, structured prediction responses, and SQL-backed data models.
- Built reproducible Python data pipelines for cleaning, transformation, validation, and feature preparation, and added monitoring, logging, and experiment documentation for repeatable evaluation.

Unisphere - Enterprise RAG & LLM Workflow Platform

Python | Claude/OpenAI APIs | LangChain | LangGraph | FAISS | FastAPI | AWS

- Built and deployed a RAG and semantic search system on AWS spanning document ingestion, chunking, embedding generation, FAISS vector indexing, semantic retrieval, and grounded LLM responses.
- Developed LangChain and LangGraph workflows with prompt design, tool/API calling, structured outputs, and intermediate reasoning steps, exposed through FastAPI REST/JSON endpoints.
- Established a retrieval evaluation loop measuring correctness, latency, answer quality, and retrieval consistency, using results to tune chunking strategy, prompt instructions, and retrieval behavior.

EDUCATION

M.S. Computer Science, University of Massachusetts Boston	GPA 3.6/4.0
B.Tech Computer Science & Engineering, VIT-AP University	CGPA 8.2/10

PUBLICATIONS & CERTIFICATIONS

BERT-QA: Question Answering with NLP & Entity Recognition	Published 2024
AWS Certified Machine Learning Engineer - Associate	Amazon Web Services
Google Cloud Professional Machine Learning Engineer	Google Cloud